

Toward Task-Aware Reuse of Cellular Molecular Receivers: History-Aware Prediction on Experimental FGF2–ERK Data

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Abstract—Using public experimental FGF2–ERK single-cell trajectories, we ask which information predicts a cellular receiver’s response to repeated chemical commands and whether it can shorten repeated use. In nested post hoc comparisons on a protocol unseen in training, reporter filtering reduced the 10-min-ahead error of a current-state predictor by 9.1%, a matched-feedback execution of the authors’ B3 model reduced the filtered predictor’s error by 4.5%, and explicit input-history features reduced the mechanistic predictor’s error by 1.0%, ending 8.2% below persistence. With predictor tuning restricted to the training protocols and a fixed, previously calibrated B3 reference, explicit input history added 0.1–2.0% in all six leave-one-protocol-out folds. For a fixed probe task we define readiness and, in simulation on B3, calibrate waiting rules on half of the posterior samples and evaluate them on the other half. A history-aware rule completed 7.4 min faster than the calibrated fixed wait, and was faster in all 20 repeated splits, but a smoothed reporter did as well. In this B3 setting the gain reflects filtering of noisy observations; no result concerns measured task reliability.

Index Terms—Molecular communication, cellular receivers, mechanism-informed prediction, receiver reuse, waiting rules, FGF2, ERK.

I. INTRODUCTION

Molecular communication (MC) with living receivers asks a chemical command to produce a specified response in a cell. Reception then means executing the requested response on time, and a previous command can change the next response through delivery persistence, receptor dynamics, intracellular feedback, or cellular state. For repeated tasks the operational question is when the receiver can be used again at matched reliability. We ground this question in recorded biology: FGF2 stimulation of PC-12 cells with ERK activity read by a FRET reporter [1], treated as a continuous receiver output without retroactive symbols or cell-fate labels.

Two lines of work bound the claim. In cell dynamics, the FGF2 data were used to infer signaling architecture, including feedback and a nontrivial washout response [1]; stimulation frequency shapes ERK responses [2]; automated chemical feedback has controlled mammalian signaling [3]; and a learned dynamical model enabled model predictive control of thousands of single cells [4]. In MC, semantic systems learn task-relevant representations [5], [6], interpretable reaction–diffusion learning has been proposed [7], experimental yeast pheromone links characterize biological receiver noise [8],

genetic-circuit receivers are designed computationally [9], calcium signaling with refractory behavior has been controlled [10], error control has been adapted to cellular noise [11], and long-term experimental MC exposes reuse tradeoffs [12]. Prediction, feedback and predictive control are therefore not new; the open question is which information makes a *next task* predictable and how much a waiting decision gains from it.

This paper makes two bounded contributions. (i) A nested comparison quantifies the contributions of reporter filtering, a calibrated mechanistic increment from the authors’ B3 model, and explicit input-history features to experimental FGF2–ERK forecasting, with phase-wise errors and leave-one-protocol-out (LOPO) validation in which predictor tuning is restricted to the training protocols. (ii) For a fixed probe task we define readiness and compare calibrated waiting rules in simulation on B3, with a smoothed-reporter baseline, a noise ablation and repeated calibration/evaluation splits. Prediction and waiting gains are evaluated separately, and no result concerns measured task reliability.

II. SYSTEM AND TASK MODEL

Let $u(t) \geq 0$ denote the commanded FGF2 concentration in ng/ml. Delivery to a receiver is represented abstractly as

$$\begin{aligned} c_R(t) &= \mathcal{T}_\theta[u](t), & \dot{x}_j &= f_{\theta_j}(x_j, c_R), \\ z_j &= g(x_j), & y_{j,i} &= z_j(t_i) + \nu_{j,i}, \end{aligned} \quad (1)$$

where c_R is the local concentration, x_j the unobserved state of cell j , z_j its noise-free normalized reporter, and $y_{j,i}$ the observation with noise $\nu_{j,i}$ at frame t_i . The operator $\mathcal{T}_\theta$ can include delivery lag and washout; neither it nor c_R is calibrated here, and the FRET ratio is not a molecular count or a cell-fate measurement. The available information $\mathcal{H}_t$ contains the executed commands and the observations whose feedback has arrived by t ; offline, each observation is assumed available at its recorded time.

We fix the amplitude and duration of the next command and decide only when to send it (Fig. 1). After a previous command ending at time 0, a 5-min probe at the same amplitude is sent at $\tau \in \mathcal{T} = \{0, 2, \dots, 120\}$ min. With z_j^τ the noise-free response when the probe is sent at τ after the recorded history,

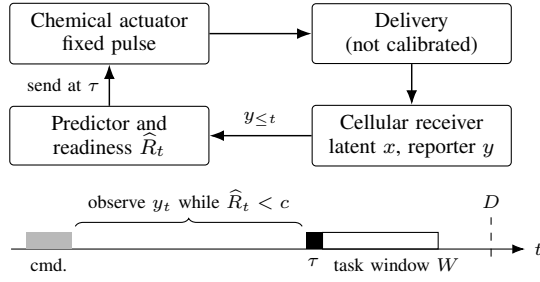


Fig. 1. System and one reuse task. The completed analysis predicts the reporter from commands and observations; delivery and latent states are not calibrated. The waiting rule is evaluated only in simulation on B3 (Sec. V-D).

and $r_{\text{naive},j}$ the same receiver's rise to the probe without a previous command, the task outcome and readiness are

$$S_j(\tau) = \mathbf{1}_{\left\{ \max_{s \in [\tau, \tau+W]} z_j^\tau(s) - z_j^\tau(\tau) \geq \kappa r_{\text{naive},j} \right\}}, \quad (2)$$

$$R_{j,t} = \mathbb{P}(S_j(t) = 1 \mid \mathcal{H}_t),$$

with task window $W = 20$ min and $\kappa = 0.5$. Observation noise affects the feedback used for decision-making; task success is scored from the noise-free model response, and decisions use only observations available by τ . The counterfactual $r_{\text{naive},j}$ is a simulation benchmark; an experiment would have to measure or estimate it, or replace it by a fixed measurable threshold. The completion time is $T = \tau + W$ on success and the deadline penalty $D = 140$ min on failure; the design objective is to minimize $\mathbb{E}[T]$ subject to $\mathbb{P}(S = 1) \geq q = 0.9$, an empirical target rather than a coverage guarantee. Readiness is task-specific: a reporter need not return to baseline before the receiver is ready for this task. The experimental data have no task labels, so Eq. (2) is evaluated only in simulation.

III. MECHANISM-INFORMED PREDICTION AND WAITING

A. Nested Predictors

Separate predictors are fit for horizons $h = 2$ and 10 min after a fixed 10-min observed warmup. Every learned predictor receives the same known future command (its exposure in 2-min bins, the concentration at $t + h$, and h); future reporters are targets only. The current-state set $\mathcal{C}$ contains $u_t, u_t^2, y_t, y_t^2, u_t y_t$, the slope $(y_t - y_{t-2})/2$ and the future-command features. Causal filters with scales $\tau \in \{2, 10, 60\}$ min summarize input and reporter history,

$$q_\tau(t) = \frac{1}{\tau} \int_0^t e^{-(t-s)/\tau} u(s) ds, \quad (3)$$

$$r_{\tau,i} = e^{-\Delta_i/\tau} r_{\tau,i-1} + (1 - e^{-\Delta_i/\tau}) y_i,$$

with $r_{\tau,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$; the scales are computational choices, not identified rate constants. O adds the reporter filters r_τ to C. As a mechanism we execute the authors' B3 model with their 1,000 posterior parameter samples. With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the samples are weighted by each cell's past reporter,

$$w_s(t) \propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \quad (4)$$

$$\hat{\delta}_{t,h} = \sum_s w_s(t) [f_s(t+h) - f_s(t)],$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors. M adds $\hat{\delta}_{t,h}$ to O, and H adds the input filters q_τ and the cumulative exposure $\int_0^t u(s) ds$ to M. The B3 increment already uses the command history and past observations, so H tests only the additional predictive value of explicit input-history features. Persistence P ($\hat{y}_{t+h} = y_t$) needs no training. Each set is fit by ridge regression with equal weight per condition and standardization from training rows; the penalty (from 10^{-8} to 10^4) and σ (seven values) are selected on validation. Adding the clock t, t^2 to each set is a sensitivity analysis. The B3 samples express population-mean uncertainty, not cell heterogeneity.

B. Calibrated Waiting Rules

A waiting rule probes at

$$\tau = \begin{cases} \inf\{t \in \mathcal{T} : \hat{R}_t \geq c\}, & \text{if such a } t \text{ exists,} \\ \max \mathcal{T}, & \text{otherwise.} \end{cases} \quad (5)$$

Three estimators of R_t are compared. The current estimator uses only y_t , through a Gaussian kernel over the (reporter, success) states of reference receivers; the smoothed estimator applies the same kernel to a causal exponential smoothing of the observations (scale 2–16 min); the history-aware estimator weights reference receivers under the known command history by all observations so far, as in Eq. (4). Per-frame probability calibration does not guarantee the reliability of an adaptive stopping rule, so the whole rule is calibrated on calibration receivers and evaluated on separate evaluation receivers. For the current and history estimators we select the first threshold on the prespecified grid $\{0.5, \dots, 0.99\}$ whose calibration success reaches q ; this is a threshold-selection heuristic guided by the completion-time objective. For each smoothing scale we first select a feasible threshold in the same way, and among feasible scales we choose the one with the smallest calibration mean completion time. If no threshold is feasible, the largest is used and the rule is reported as missing the target. The comparator fixed wait is the shortest grid wait whose calibration success reaches q . As a sensitivity analysis, the objective-aligned choice selects, among all feasible thresholds (or scale–threshold pairs, or waits), the one with the smallest calibration mean completion.

IV. DATA AND EVALUATION PROTOCOL

A. Data and Roles

All numerical files come from the paper-linked `Mijan/LFNS_MSB` repository at pinned commit `5c917ab` [13], with the associated data record [14]; every file matches its Git blob identifier. Commands follow the authors' model XML with half-open pulses; mixed pulses span $[1, 4)$, $[24, 54)$ and $[114, 119)$ min, and for this protocol the 10–30 min per-cell normalization was reconstructed from unnormalized exports. Table I lists the 22 conditions. An earlier pilot tested on the viewed mixed blocks, which now serve only as context. The roles were frozen, with a SHA-256 record,

TABLE I
PROTOCOLS, ROLES AND SINGLE-CELL TRAJECTORIES (22 CONDITIONS,
1,565 CELLS, 2-MIN FRAMES). B3 USE: CONCENTRATIONS WHOSE MEANS
THE AUTHORS FITTED (FIT) OR USED IN THEIR ARCHITECTURE
COMPARISON (ARCH.).

Protocol	Role	ng/ml	Cells	B3 use
Sustained	Train	0.25, 2.5, 25, 250	248	fit 2.5, 250
3/20 pulses	Train	2.5, 25, 250	207	fit 2.5, 250
Single 5 min	Val.	0.25, 2.5, 25, 250	370	arch. 2.5, 250
Single 10 min	Test A	0.25, 2.5, 25, 250	248	—
Single 60 min	Test B	2.5, 25, 250	167	—
Mixed	Test C	0.25, 25	171	—
Mixed	Viewed	2.5, 250	154	arch. 2.5, 250

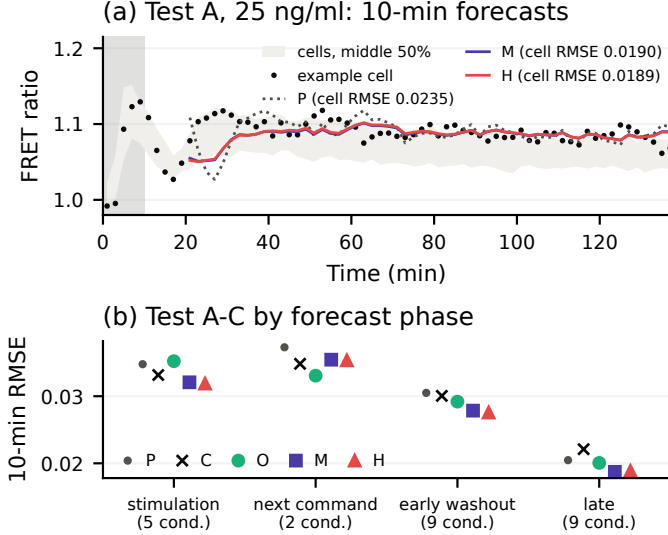


Fig. 2. Experimental 10-min prediction. (a) Test A at 25 ng/ml: the cell with median H error (of 64), its observations, the middle 50% of cells, and the forecasts of P, M and H at their target times; gray: command. (b) Condition-equal RMSE on Tests A–C by forecast phase (axis not starting at zero).

before any newly used response file was downloaded. Test A is a new protocol. Test B’s $[0, 60]$ timing was inferred from the authors’ B3 simulations, with a ± 1 min sensitivity, and later confirmed by their simulation summary. Test C uses new concentrations recorded in the session of the viewed blocks. The nested comparison and the waiting studies were each frozen in their own protocol before running, but all are post hoc with respect to inspected data.

B. Mechanism, Validation and Scores

B3 is executed by parsing the authors’ model files; no equation is retyped, no parameter is refit, and our posterior-predictive mean matches their exported predictions within 8.5×10^{-6} (acceptance gate 10^{-3}). Because the authors fitted B3 to the sustained and 3/20 means and compared architectures on the single 5-min and viewed mixed blocks at 2.5 and 250 ng/ml (Table I), M and H use a calibrated reference, not a model blind to those protocols. In LOPO each of six protocol groups is held out in turn, with predictor tuning restricted to the training protocols: σ is selected on the training groups, the

penalty by inner leave-one-group-out over the training groups, and the held-out group enters no standardization, selection or fit, while B3 remains a fixed, previously calibrated reference. Only the 10- and 60-min pulse protocols were absent from the original B3 fitting and architecture comparison. Forecast rows are labeled by phase: stimulation (command on at t), next command (a pulse starts in $(t, t+h]$), early washout (last pulse ended at most 30 min before t), and late. After its warmup Test A contains only washout rows, so it mainly tests washout prediction. The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}_{\text{cond}}(h) = \left[\frac{1}{|D|} \sum_{d \in D} \frac{1}{N_{d,h}} \sum_{i \in d} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (6)$$

reported with per-condition directions. Rows and cells are not independent interventions and experimental-day identifiers are unavailable, so no significance test is reported.

V. RESULTS AND DISCUSSION

A. Nested Prediction

Table II(a) reports the nested predictors. On Test A at 10 min, reporter filtering reduced the error by 9.1% relative to C, the B3 increment by 4.5% relative to O, and explicit input-history features by 1.0% relative to M, giving 0.02098, 8.2% below persistence; these reductions use different baselines and are not additive. Because M already uses the command history and past observations, the H-versus-M gain measures only the added value of explicit input-history features, and it was small and not consistent across conditions: two of four concentrations on Test A, 0.4% and 0.1% on B and C, and 0.2% at 2 min. The mechanistic increment helped in every test role (M versus O: 9.8% on B and 6.0% on C), whereas the filtering gain vanished on B and C. Clock terms did not help (0.1–2.4% worse on the tests), and C alone was worse than persistence on Tests A and B.

B. Forecast Phases

In Fig. 2(a) persistence lags the washout by the horizon, whereas M and H follow it (cell RMSE 0.0235, 0.0190 and 0.0189). By phase (Fig. 2(b)), the mechanistic increment reduced the error of O in every phase except the next-command window (by 8.9% during stimulation, 4.6% in early washout and 6.6% late); in early washout O, M and H reached 0.02919, 0.02785 and 0.02757. During stimulation the reporter filters hurt (O 0.03522 versus C 0.03316), and in the late phase M was best with H slightly worse. The next-command window is covered by two mixed conditions with ten forecast origins each. The B3 increment helped at 0.25 ng/ml (O 0.03652, M 0.03485, H 0.03486) but raised the error at 25 ng/ml (0.02920, 0.03603 and 0.03581), so the pooled error rose from 0.03306 to 0.03544. Across the two mixed conditions contributing to this window, the B3 increment thus increased the aggregate forecast error, which limits its present use for predicting responses to an imminent command despite its benefits during washout.

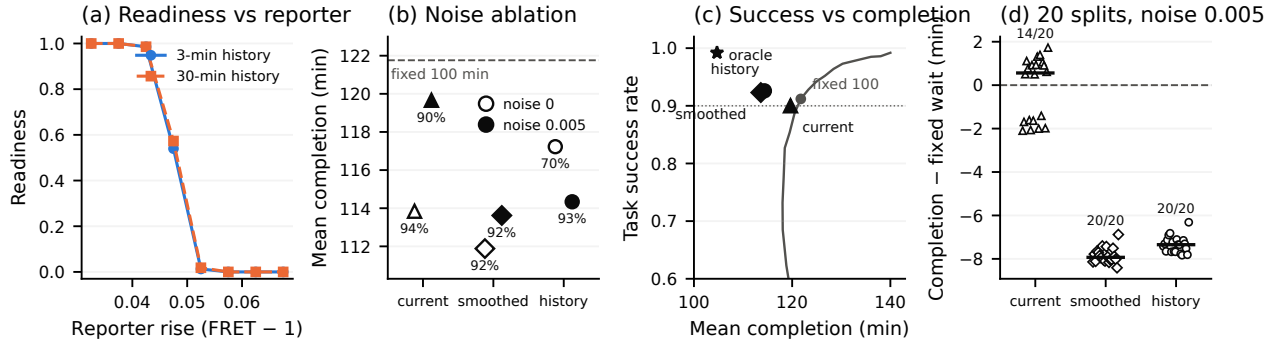


Fig. 3. Calibrated waiting on the authors' B3 model (25 ng/ml; task: probe rise $\geq 50\%$ of the naive rise within 20 min). (a) Readiness versus the current noise-free reporter after 3- or 30-min commands. (b) Mean failure-penalized completion on evaluation samples of the primary split at observation noise 0 (open) and 0.005 (filled), labeled with success; dashed: calibrated fixed wait. (c) Success versus mean completion at noise 0.005; gray: fixed waits; dotted: target 0.9. (d) Twenty calibration/evaluation splits with fresh observation noise: completion minus that of the calibrated fixed wait (bars: medians; numbers: splits whose evaluation success reached 0.9). "History": the history-aware rule. Thresholds and the fixed wait were chosen on calibration samples. Simulation of one model structure, not measured cells.

TABLE II

CONDITION-EQUAL 10-MIN RMSE OF NESTED PREDICTORS UNLESS NOTED. (A) VAL.: SINGLE 5-MIN PULSE; A: SINGLE 10-MIN PULSE; B: SINGLE 60-MIN PULSE (INFERRED TIMING); C: MIXED PULSES AT 0.25 AND 25 NG/ML. P AND C ARE PRE-REGISTERED PREDICTORS; O, M AND H ARE POST HOC. BOTTOM ROWS COUNT CONDITIONS WITH LOWER MSE. (B) CORRECTED LOPO WITH PREDICTOR TUNING RESTRICTED TO THE TRAINING PROTOCOLS; B3 PRIOR USE FROM THE AUTHORS' INFERENCE CONFIGURATION. BOLD: SMALLEST PER COLUMN (A) OR ROW (B).

(a) Fixed roles

Model	Val.	A	B	C	A, 2 min
P: persistence	0.02353	0.02287	0.02651	0.02876	0.01302
C: current + slope	0.02503	0.02442	0.02715	0.02825	0.01342
O: + reporter filters	0.02403	0.02220	0.02715	0.02827	0.01283
M: + B3 increment	0.02197	0.02120	0.02448	0.02658	0.01263
H: + input history	0.02195	0.02098	0.02439	0.02656	0.01260
H beats M	1/4	2/4	2/3	1/2	2/4
M beats O	4/4	3/4	2/3	2/2	3/4
O beats C	4/4	4/4	2/3	1/2	4/4

(b) Leave-one-protocol-out

Held-out protocol	O	M	H	B3 prior use
Sustained	0.02102	0.01969	0.01941	fit 2.5, 250
3/20 pulses	0.03680	0.02886	0.02830	fit 2.5, 250
Single 5 min	0.02243	0.02153	0.02139	arch. 2.5, 250
Single 10 min	0.02084	0.02106	0.02104	none
Single 60 min	0.02701	0.02410	0.02369	none
Mixed, 0.25/25	0.02709	0.02623	0.02610	other conc. arch.

C. Corrected Leave-One-Protocol-Out

In the corrected LOPO (Table II(b)), H beat M in all six folds, by 0.1–2.0%, and beat persistence by 8.0–30.6%. The B3 increment contributed more than reporter filtering in five folds (up to 21.6% on 3/20). Of the two protocols absent from the original B3 fitting and architecture comparison, the 60-min pulse benefited from the increment (M 10.8% below O), whereas with the 10-min pulse held out O was best and M was 1.1% worse. The sustained and 3/20 folds contain conditions used to fit B3, so their mechanistic gains are not out-of-sample

for the mechanism. An earlier LOPO that reused the primary penalties is kept in the package.

D. Calibrated Waiting on B3

Readiness and waiting were simulated on B3 with 25 ng/ml commands, a 3- or 30-min previous command, and each posterior sample in turn as the receiver, observed every 2 min with noise of s.d. 0.005. Readiness depended almost only on the current reporter for both histories (Fig. 3(a)). A first task version requiring 80% of the naive response was infeasible for every sample within 120 min and is reported in the package, and the registered candidate waits of 6–48 min never succeeded. On the primary split, with rules calibrated on 500 samples and evaluated on the other 500 (Fig. 3(c)), the history-aware rule reached 92.6% success with a mean completion of 114.3 min, faster than the calibrated fixed wait (100 min: 91.2%, 121.8 min) by 7.4 min (paired 95% bootstrap interval over evaluation samples, 6.6–8.2 min); the current-observation rule reached 90.0% at 119.7 min. A smoothed reporter (16-min scale) reached 92.3% at 113.6 min, 0.7 min faster than the history rule (0.3–1.1 min) with a success difference of 0.3 points (–1.2 to 1.8). These pooled values average the two histories, and the fixed wait hides a history dependence (Table III): 97.4% success after the 3-min command but 85.0% after the 30-min command, whereas both adaptive rules stayed between 91% and 95%. Over 20 splits with fresh observation noise (Fig. 3(d)), the history rule was faster than the fixed wait in every split (by 6.3–7.8 min), the smoothed rule was faster than the history rule in every split (by 0.2–1.0 min), and evaluation success reached 0.9 in 20 splits for both, 19 for the fixed wait and 14 for the current rule. The objective-aligned selection chose the same thresholds, scales and waits as the heuristic in all 40 split-noise combinations. At zero observation noise (Fig. 3(b)), no candidate history threshold attained the calibration target: calibration success was 67.2%, and evaluation success under the fallback threshold was 70.1% (no split reached the target), as the weights collapsed onto essentially one reference sample (median effective sample

TABLE III

CALIBRATED WAITING ON THE PRIMARY SPLIT BY PREVIOUS COMMAND (NOISE 0.005): EVALUATION SUCCESS S AND MEAN FAILURE-PENALIZED COMPLETION T AFTER THE 3- AND 30-MIN COMMANDS. POOLED VALUES IN THE TEXT ARE THE EQUAL-WEIGHT MEANS OF THE TWO HISTORIES.

Rule	S_3 (%)	T_3 (min)	S_{30} (%)	T_{30} (min)
Current obs.	88.8	115.3	91.2	124.0
Smoothed (16 min)	92.6	108.6	92.0	118.6
History-aware	91.0	109.1	94.2	119.6
Fixed 100 min	97.4	120.5	85.0	123.0
Oracle	100.0	99.0	98.4	110.4

size 1.1 versus 28.9 with noise); the current rule reached 93.6% at 113.8 min. In this B3 setting, reporter smoothing matched the history-aware rule, and the results do not establish an additional waiting benefit from latent memory; the oracle (99.2%, 104.7 min) bounds what better state estimation could add.

E. Evidence Boundaries

Seven limits qualify these results. First, in the frozen analysis of the main fresh test, history features alone improved on persistence by only 0.7%, and explicit input-history features add about 1% beyond the mechanism. Second, the nested and hybrid comparisons are post hoc on inspected roles, not independent confirmation. Third, Test C shares a session with inspected data and no experimental-day identifiers exist, so protocol generalization is not cross-day generalization. Fourth, the waiting results come from B3 simulation: success over posterior samples is not the reliability of a measured cell population, the posterior samples are not validated cell heterogeneity, and the 90% target carries no coverage guarantee. Fifth, the waiting gain is attributable to filtering in this setting; the zero-noise failure of the history rule reflects finite reference samples and the likelihood bandwidth and says nothing general about biological memory, and whether measured cells carry history that matters for control beyond the reporter remains open. Sixth, the task threshold was lowered from 80% to 50% after the first version proved infeasible. Seventh, amplitude and duration were fixed, so nothing is claimed about joint exposure–time optimization. Measured benefits require calibrated delivery, prespecified tasks, and new independent-session experiments.

VI. CONCLUSION

On experimental FGF2–ERK trajectories, reporter filtering and a calibrated mechanistic increment from the authors’ B3 model account for most of the 10-min prediction gain; explicit input-history features add a small gain that holds across protocol folds but not across conditions, and the mechanism is least reliable before an imminent command. In simulation on B3, calibrated history-aware and smoothed-reporter waiting rules met a 90% target faster than fixed waiting, with no additional benefit from the history model.

DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [1], [13], [14]; author files keep their attribution and receive no new license.

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